INCH-POUND

MIL-W-63451 (AR)
AMENDMENT 5
18 October 1996
SUPERSEDING
AMENDMENT 4
18 June 1986

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E2
LINER FOR

This Amendment forms a part of Military Specification MIL-W-63451 (AR) dated 17 June 1981 and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

In title add “and M207E5” after “M207E2”

1.1, add, “and M207E5” after M207E2” and change “warhead” to warheads”.

PAGE 2

Add 2.2 as follows:

“2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.”

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM B 152	Copper, strip and Rolled Bar
ASTM B 248	General Requirements for Wrought Copper and Copper - Alloy Plate, Sheet, Strip, and Rolled Bar

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102)

AMSC N/A FSC 1336

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Add 3.6, 3.6.1 and 3.6.2 as follows:

"3.6 Process control

3.6.1 Heat treat process documentation. Concurrence with submission of the First Article Sample (see paragraph 4.3), the contractor shall submit a written description of the heat treatment process used to produce the samples together with the means by which the process can be verified in production. The method of verification shall be reviewed and approved by the government technical agency as required in paragraph 6.4. See Data/Item Description DI-P-1604 (Tailored). Proposed changes to the process which produced the accepted First Article Sample shall be submitted to the government prior to actual implementation of the change. A new First Article Sample will be required for each process change.

3.6.2 Liner traceability. The contractor shall also submit a method to establish liner traceability with specific annealed batches. This information shall be furnished as a part of DI-P-1604 (Tailored) above and shall be reviewed by and must be acceptable to the government.

* Add new paragraph 3.7 as follows:

" 3.7. Static Penetration Test. A total of ten (10) finished machined liners shall be delivered to the designed government facility for loading into warheads and testing in accordance with paragraph 6.4. This test is for information only."

PAGE 3

4.3.1 Under "Liner" add the following:

" Sample material of the starting stock shall be selected prior to fabrication and used for hardness, chemical composition and grain size tests. In addition, ten finish machined liners from a single annealed batch shall be used for testing of grain size and hardness."

* Add the following new paragraph:

"4.3.4 Completion. Upon completion of the first article test, all samples and components shall be send to the government technical agency as required in paragraph 6.4. After completion of the first article test, written authorization to start liner production shall be obtained

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from the government technical agency as listed in paragraph 6.4."

PAGE 4

Delete Page 4 and substitute page 4 included.

* Table 1, First article inspection, add new test as follows:

" Examination/ Test	No. of Sample Units	Req't Para.	Para. Ref./ Insp. Method "
Penetration Test	10	3.7	4.6

PAGES 6 & 7 & 7a & 7b

Delete pages 6 and 7 and substitute pages 6, 7, 7a and 7b included.

PAGE 8

Add 4.4.3.1 & 4.4.3.1.1 as follows:

"4.4.3.1 Heat treatment. Tests shall be performed in accordance with 4.5.2.4 and 4.5.2.5."

* 4.4.3.1.1 Heat treatment. Sample testing shall be in accordance with the following:

a. Randomly select liners from annealed batch per table II, Level VI, and perform tests in accordance with paragraph 4.5.2.4 and 4.5.2.5. An annealed batch shall consist of the quantity of liners that has been annealed in the same heat treatment equipment by the same method in one unchanged process at the same time and temperature. Batches presented by the unacceptable liners shall be rejected.

b. In additional to the test results, evidence must be submitted to establish that the heat treat process for each month's production is unchanged from the approved process (see paragraph 3.6.1). This evidence shall be in the form of permanent hard copy records, such as temperature-time tracings."

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Liner	SHEET 1 OF 1		DRAWING NUMBER See Below
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Liner, (Dwg. 9340752)			
	Examination for Defects	20	3.2	4.4.2.1
	Hardness of starting stock	Sample Material	3.2	4.5.2.1
	Chemical composition of starting stock	Sample Material	3.2	4.5.2.2
	Grain size of starting stock	Sample Material	3.2	4.5.2.3
	Hardness of finish liner (See Note 1)	10	3.2	4.5.2.4
	Grain size of finish liner (See Note 1)	10	3.2	4.5.2.5

NOTES:
1. The same 10 Liners are used for both tests.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 3		DRAWING NUMBER
4.4.2.1	Liner			9340752
				NEXT HIGHER ASSEMBLY 9340753
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Run out of O.D. with Datum A & B			
102	Outside Diameter (O.D)	0.40%	3.2	Gage
103	Inside Diameter (I.D)	0.40%	3.2	Gage
104	Length of Liner step	0.40%	3.2	Gage
105	Length from base to datum B	0.40%	3.2	Gage
106	Straightness on outside surface	0.25%	3.2	Gage
107	Runout of base with datum A & B	0.25%	3.2	Gage
108	Length from datum A to B	0.40%	3.2	Gage
109	Length from datum B to inside spherical radius	0.40%	3.2	Gage
110	Wall thickness at apex end	0.40%	3.2	Gage
111	Wall thickness at larger basic dimension at apex end	0.25%	3.2	Gage
112	Wall thickness at small basic dimension, apex end	0.25%	3.2	Gage

NOTES:
1. Measurements shall be taken on both 42 degrees and 28 degrees angle surface.
2. Data Item Description DI-T-5247 (Tailored) applies for these tests.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner	2 OF 3	9340752
			NEXT HIGHER ASSEMBLY 9240753

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
113	Length from larger to small basic dimension at apex end	0.40%	3.2	Gage
114	Runout between two basic dimension on outside surface at apex end (See Note 8 of dwg.)	0.25%	3.2	Gage
115	Length from datum B to basic dimension at apex end	0.40%	3.2	Gage
116	Wall thickness variation in any axial plane (See Note 4 of dwg.)	0.25%	3.2	Gage
117	Wall thickness variation in any transverse plane (See Note 2)	100%	3.2	Gage
118	Wall thickness in area at 3.639 dimension	0.40%	3.2	Gage
119	Evidence of discontinuity of machined surface	100%	3.2	Visual
120	Hardness of starting stock (See Note 3)	Material Sample	3.2	4.5.2.1
121	Chemical composition of starting stock (See Note 3)	Material Sample	3.2	4.5.2.2

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner	3 OF 3	9340752
			NEXT HIGHER ASSEMBLY 9240753

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
122	Grain size of starting stock (See Note 3)	Material Sample	3.2	4.5.2.3
123	Hardness of finished liner (See Note 3)	4.4.3.1 4.4.3.1.1.	3.2	4.5.2.4
124	Grain size of finished liner (See Note 3)	4.4.3.1 4.4.3.1.1.	3.2	4.5.2.5
MINOR				
201	Surface finish improper	0.65%	3.2	Gage
202	Inside spherical radius, apex end (See Note 1)		3.2	Gage
203	Outside spherical radius, apex end	0.65%	3.2	Gage
204	Present of 4 slots on O.D.	0.65%	3.2	Visual
205	Evidence of poor workmanship	1.00%	3.5	Visual

1. All mew and resharpened tooling shall be certified prior to use. A cast of the inside spherical radius of the liner shall be made from "dental plaster" or equivalent, which shall be inspected by an optical comparator of 10x magnification or greater. The inspection shall be performed at the beginning of the shift half way through the shift and whenever tooling is changed.

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* Add Table II as follows:

“ Table II - Attributes sampling inspection

ANNEALED BATCH SIZE	INSPECTION LEVEL		
	IV	V	VI
26 to 90	13	5	3
151 to 500	32	20	8
501 to 1200	50	20	13
1201 to 3200	50	32	13

Number under inspection level indicate sample size, accept on zero and reject on one or more for all inspection level. “

Add 4.5.2, 4.5.2.1, 4.5.2.2, 4.5.2.3, 4.5.2.4 and 4.5.2.5 as follows:

“4.5.2 Metallurgical & chemical tests. The following tests shall be performed using equipment and procedures prescribed by ASTM B248.

4.5.2.1 Hardness of starting stock. The Rockwell hardness of a sample of starting stock shall be evaluated. These test data shall be independent from the supplier data. Failure of the sample to comply with the requirements of ATM B152 shall cause rejection of that quantity of that starting stock represented by the sample. The hardness test data shall be submitted in accordance with DI-T-5247. (Tailored) (See 6.4)

4.5.2.2 Chemical composition of starting stock. Sample of starting stock shall be analyzed for chemical requirements. These test data shall be independent from supplier data. Failure of the sample to comply with the requirements of ASTM B152 shall cause rejection of the quantity of starting stock represented by the sample. Result of these tests shall be submitted in accordance with DI-T-5247 (Tailored) (/see 6.4).

4.5.2.3 Grain size of starting stock. Sample of starting stock shall be tested for grain size. This test shall be independent from supplier data. Failure of the sample to comply with the requirements of ASTM B152 shall cause

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rejection of that quantity of starting stock represented by the sample. The grain size test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4).

4.5.2.4 Hardness of finished liner. The sample liners shall be selected longitudinally into four equal parts and marked for exact location relative to the finished liner. The Rockwell F hardness readings shall be taken one inch from the base and every one inch thereafter in a longitudinally direction, from base to apex, for seven readings on each section. (See Fig. 1) A total of twenty-eight readings shall be taken for the finished liner. The hardness readings shall be identified as to their location on the liner. The recorded readings shall be analyzed for variations in hardness in both the longitudinal and circumferential directions. If any of the sample liners do not meet the hardness requirement specified on the drawing, the liner heat treat batch represented by the samples shall be rejected. The hardness test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4). This is a destructive test.

4.5.2.5 Grain size of finished liner. A test section from 4.5.2.4 shall be utilized for taking grain size. Grain size shall be taken on one of the quartered sections in a longitudinal 1½ inch from the apex and 1½ inch apart from each other for a total of four readings per sample liner. (See Fig. 1) If any of the sample liners do not meet the grain size requirements as specified on the drawing, the liner heat treat batch represented by the sample shall be rejected. The grain size test data shall be submitted in accordance with I-T-5247 (Tailored) (See 6.4) “

* Add paragraph 4.6 as follows:

“4.6 Static penetration test. The ten (10) liners shall be press loaded into ten (10) warheads in accordance with Drawing Number 9399750 and shall be in accordance with MIL-W-48689 (AR).”

6.1, add “and M207E5” after “M207E2”.

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Deleted 6.4 in its entirety and substitute with the following:

"6.4 Data Item requirements. Delivery data required by the specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3	Ammunition Data	DI-L-1410
3.6, 3.6.1, 3.6.2	Manufacturing-Methods Report	DI-P-1604 (Tailored)
3.7	Test Report	DI-T-5247 (Tailored)

* Delete the sentence " The above data items …… the following:" and substitute with " The above data item shall be delivered to the following government technical agency:"

* Change the address from " Commander …… 5001 " to :

" Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-QAC-S, Bldg 62 North, Picatinny Arsenal, NJ 07806-5000"

PAGE 10

Add Figure 1, included.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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SCHEMATIC OF SECTIONED LINER

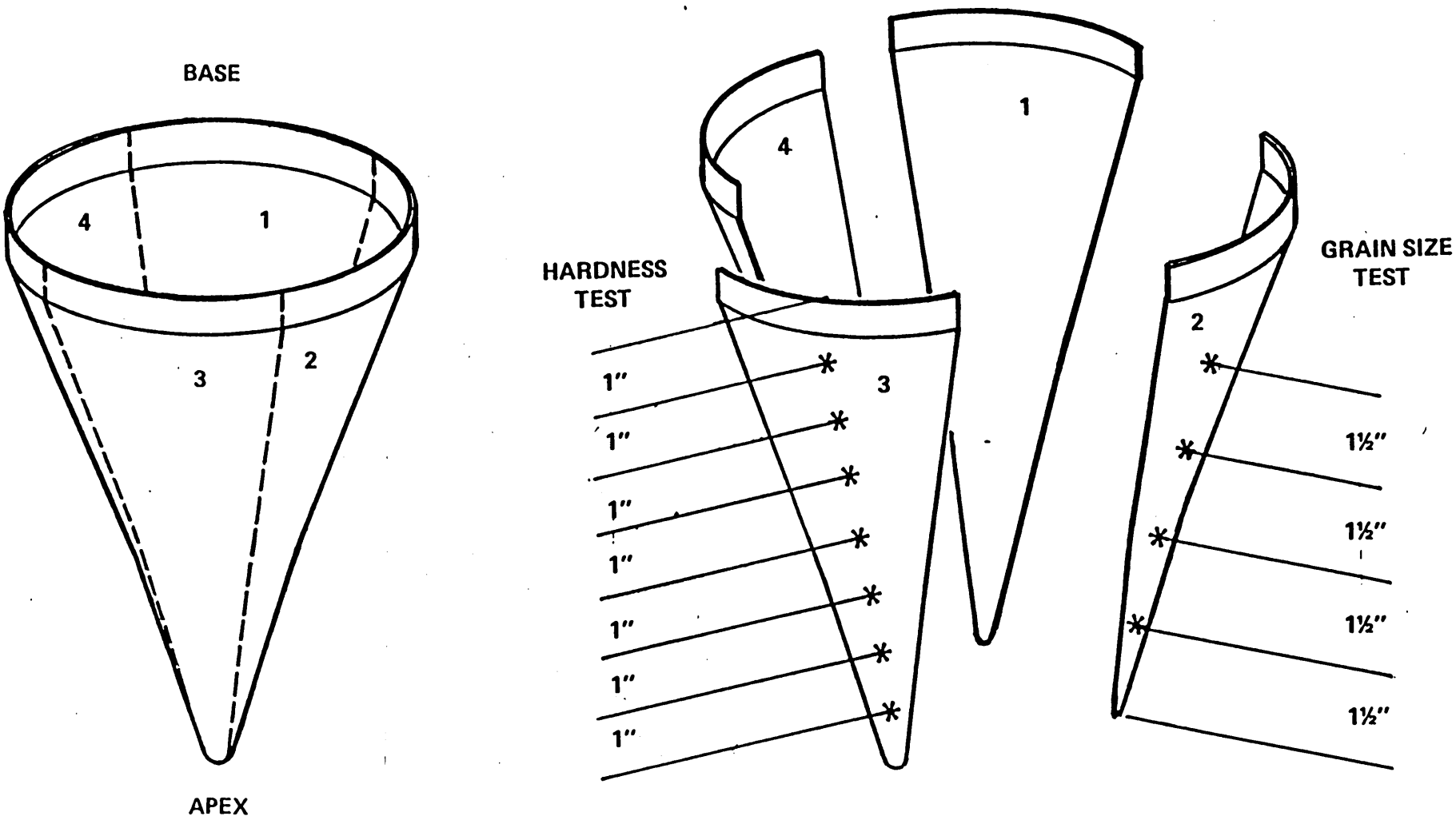


FIGURE 1

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